Area/regs

Retiming trades register area for shorter clock period , same function, different register placement

The idea

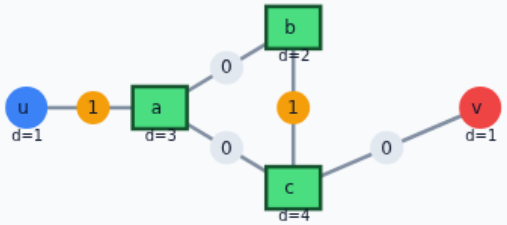
- Forward on a

Registers ≈ area

AREA / REGISTERS

Registers ≈ area

Step 1 / 5 · proxy · module03-05-area-registers



Explanation

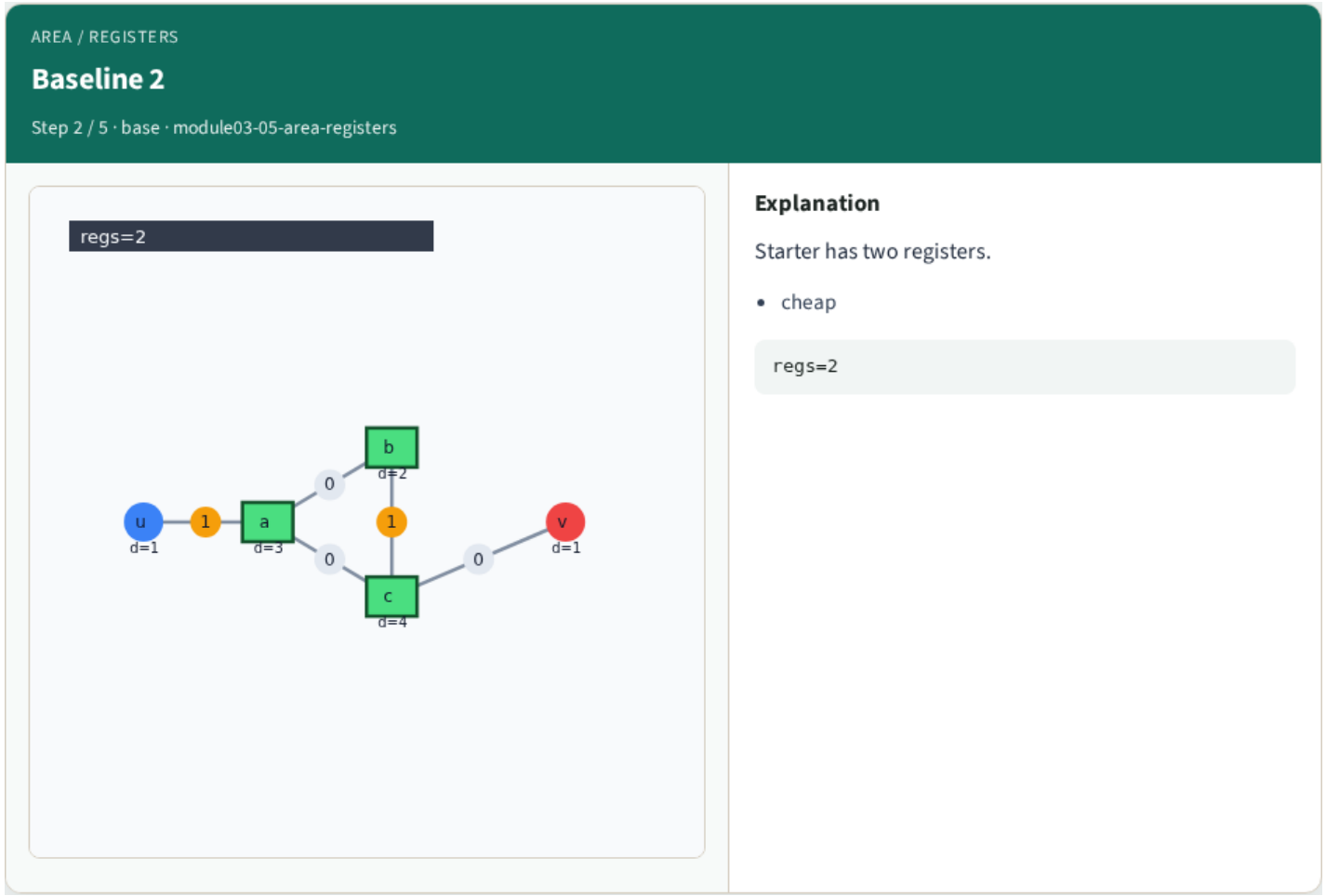
Total w is a stand-in for flip-flop area.

- FF count

```
sum w
```

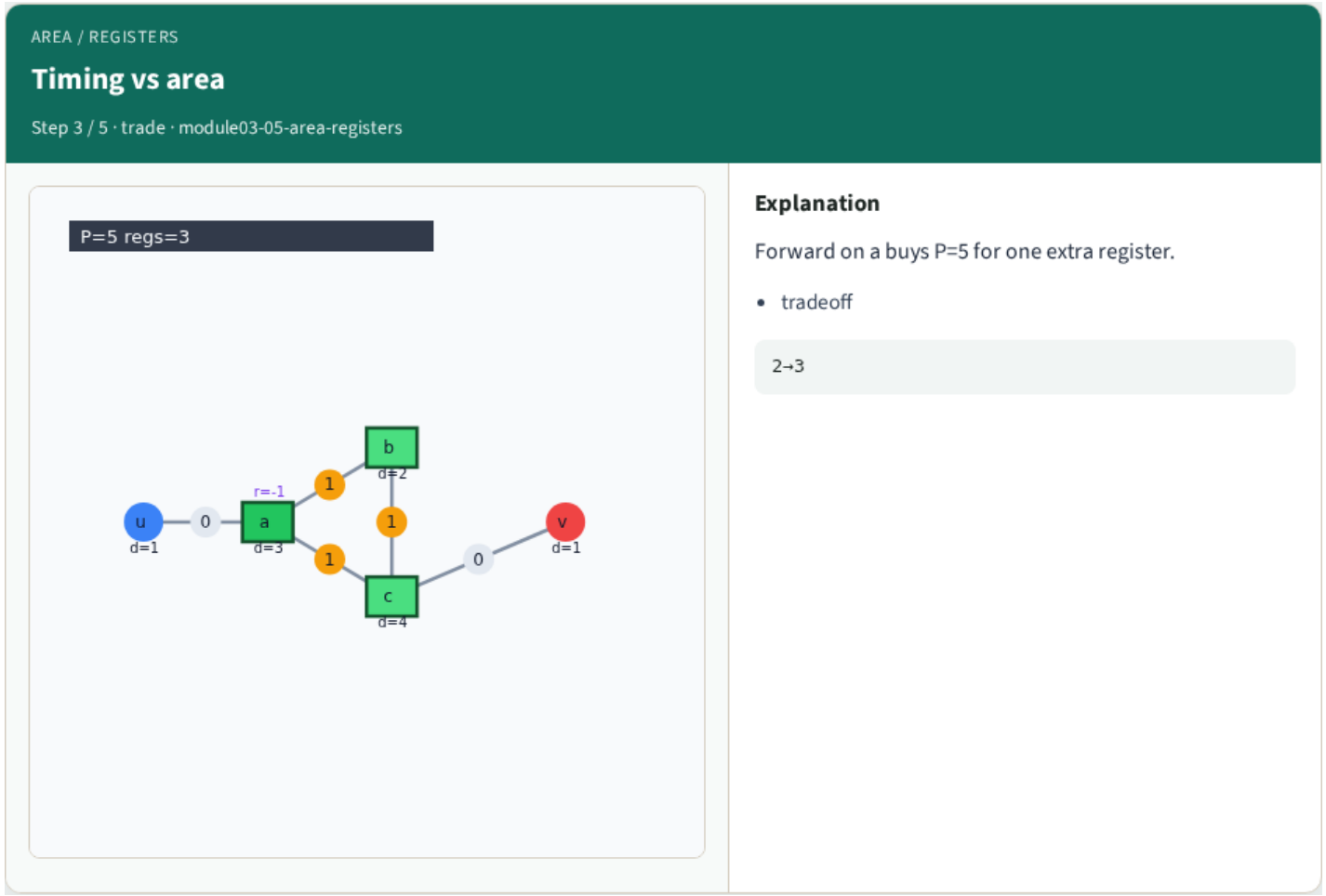
Registers ≈ area

Baseline 2



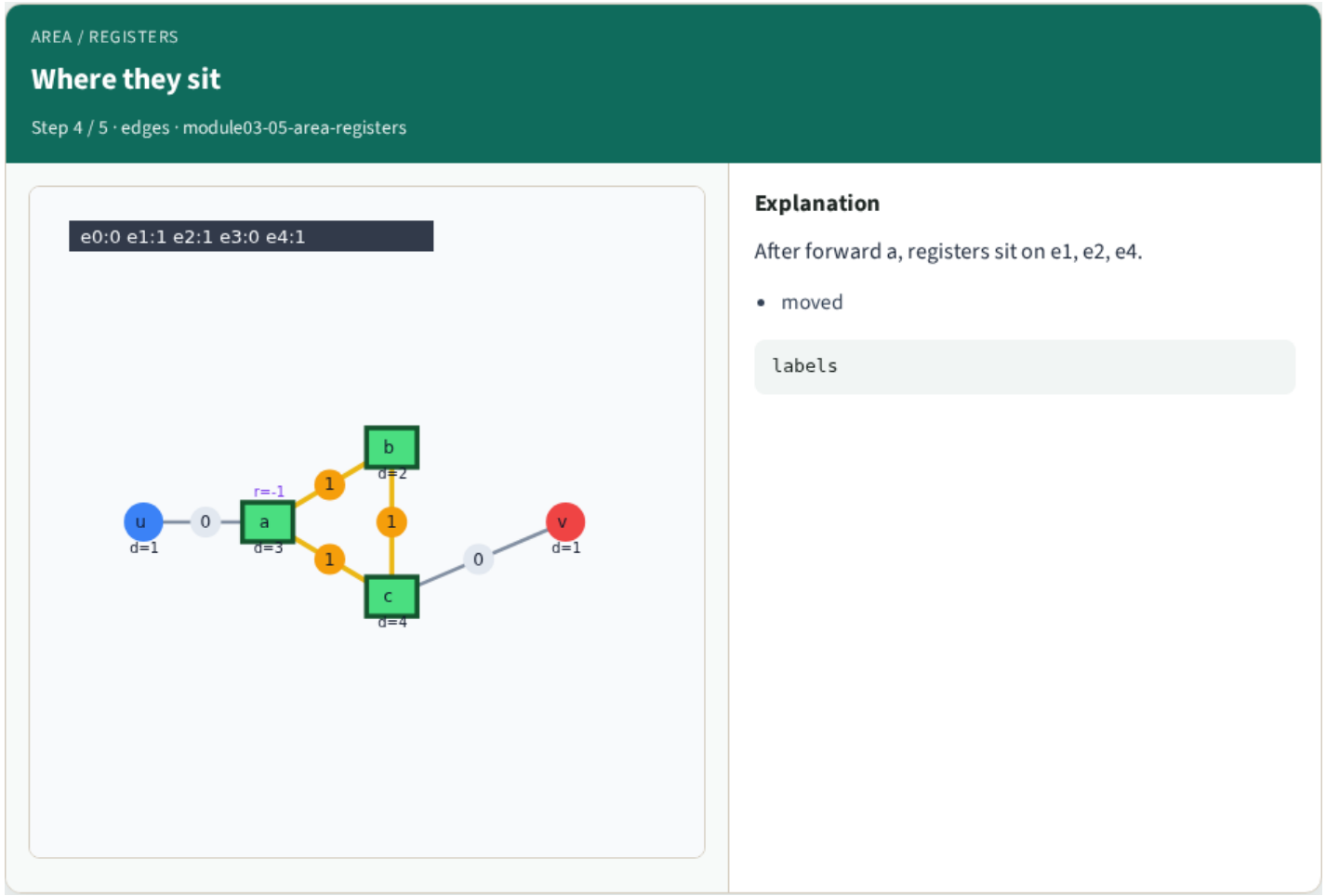
Baseline 2

Timing vs area



Timing vs area

Where they sit



Where they sit

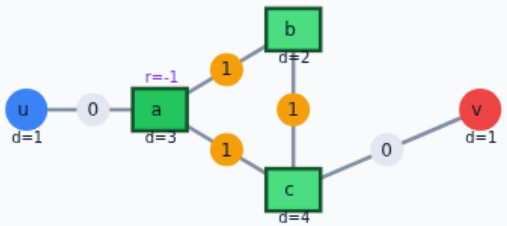
Done

AREA / REGISTERS

Done

Step 5 / 5 · done · module03-05-area-registers

P=5 regs=3



Explanation

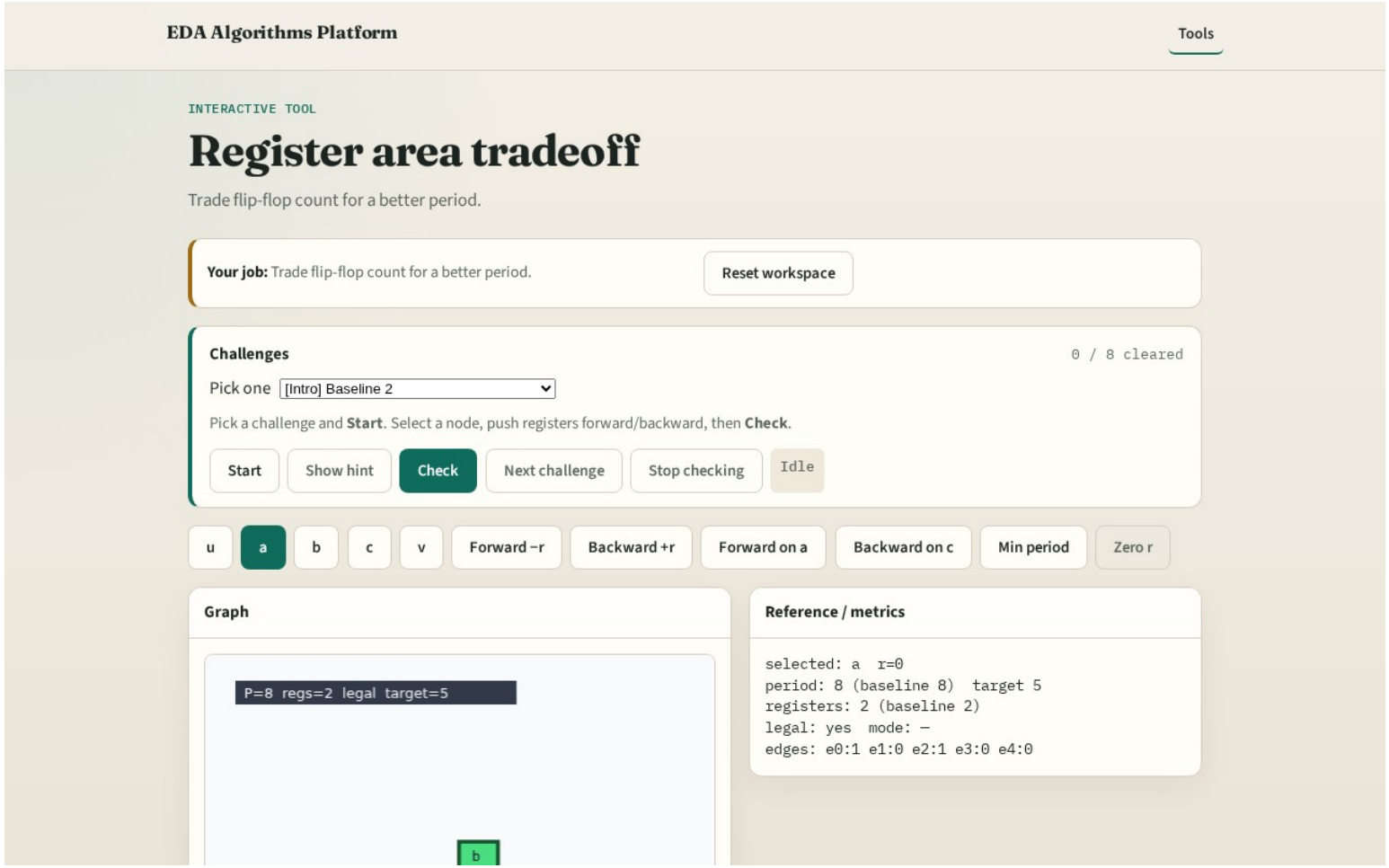
Next: offline compare, then wrap toward equivalence or STA.

- learn_equivalence

```
Wrap
```

Done

Browser lab track



Browser lab starter

Implement track

- Print register_count at r=0 and after forward on a (2 vs 3)

Pitfalls

- Assuming retiming always reduces registers
- Ignoring period when counting area

Your turn

- Next: offline benchmark, then wrap